

Final Exam Answers

Exercise 1:

- ① False: In $\mathbb{Z}/2024\mathbb{Z}$, We have $\overline{2} \neq \overline{0}$ and $\overline{1012} \neq \overline{0}$ but $\overline{2} \cdot \overline{1012} = \overline{2024} = \overline{0}$. Then $\mathbb{Z}/2024\mathbb{Z}$ contains as a ring zero divisors.
- ② False: In $\mathbb{R}[x]$, x^4+1 does not admit roots in $\mathbb{R}$ but it is reducible, indeed, $x^4+1 = (x^2+ix+1)(x^2-ix+1)$.
- ③ True: In $\mathbb{Q}[x]$, x^3-2 is irreducible, since the reducibility of x^3-2 because its degree is 3 is equivalent to x^3-2 admits a root in $\mathbb{Q}$, which is equivalent to $\sqrt[3]{2} \in \mathbb{Q}$, which is impossible.
- ④ True: since x^3+x+1 is of degree 3, the reducibility is equivalent to x^3+x+1 admits a root in $\mathbb{Z}/5\mathbb{Z}$, which is not the case. $p(\overline{0})=\overline{1}$, $p(\overline{1})=\overline{3}$, $p(\overline{2})=\overline{1}$, $p(\overline{3})=\overline{1}$, $p(\overline{4})=\overline{4}$.
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⑤ True: $f: \mathbb{C} \rightarrow \mathbb{C}$ a field homomorphism

$$f(n) = n \quad \forall n \in \mathbb{R}, \text{ then } f(a+ib) = f(a) + f(i)f(b)$$

$$\forall a, b \in \mathbb{R}, \text{ so } f(a+ib) = a + f(i)b.$$

$$\text{on the other hand } i^2 = -1 \text{ then } f(i^2) = f(-1)$$

$$\text{then } f^2(i) = f(-1) = -1, \text{ so } f(i) = \pm i$$

$$\text{Conclusion } f: \mathbb{C} \longrightarrow \mathbb{C} \text{ or } f: \mathbb{C} \longrightarrow \overline{\mathbb{C}}$$
$$z \longrightarrow z \quad \quad z \longrightarrow \overline{z}$$

⑥ True: Let x be a zero divisor, then $x \neq 0$ and there exists $y \neq 0$ such that $xy = 0$. If we suppose x invertible and x^{-1} is its inverse,

We get

$$x^{-1}xy = 0 \text{ i.e.,}$$

$$1 \cdot y = 0 \text{ then } y = 0 \text{ which}$$

is a Contradiction.

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Exercise 2 a) $P_n = (x+1)^n - x^n - 1 = \sum_{i=0}^n a_i \cdot x^i$

$$P_n = \sum_{k=0}^n C_n^k x^{n-k} - x^n - 1$$

$$= \underbrace{C_n^0 x^n}_{1} + C_n^1 x^{n-1} + \dots + \underbrace{C_n^n x^0}_{1} - x^n - 1$$

$$= C_n^1 x^{n-1} + \dots + C_n^{n-1} x$$

Then $\deg P = n-1$, $\therefore$ the leading coefficient is C_n^1 . The constant term is 0.

b)

$$P_n(j) = (j+1)^n - j^n - 1, \text{ knowing that } j^3 = 1 \text{ and } 1+j+j^2 = 0 \text{ we obtain}$$

$$P_n(j) = (-j^2)^n - j^n - 1 = (-1)^n j^{2n} - j^n - 1 = 0 \text{ if and only if } n \equiv 1, 5 \pmod{6}$$

c) $P_n = x(x+1)Q + ax + b \quad a, b \in \mathbb{R}.$

$$P_n(0) = 0, \text{ then } b = 0.$$

$$P_n(-1) = -(-1)^n - 1 = (-1)^{n+1} - 1 = -a + b, \text{ then}$$

$$a = (-1)^n + 1, \text{ so } P_n = x(x+1)Q + ((-1)^n + 1)x$$

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2) a) It is clear that $P_7(0) = 0$ and $P_7(-1) = 0$
 then the integer roots of P_7 are 0, and -1.

b) $P_7(j) = 0$ (from 1.b).

$$P_7' = 7(x+1)^6 - 7x^6, \quad P_7'(j) = 7(-j^2)^6 - 7j^6 = 7(j^{12} - j^6) = 0.$$

$$P_7'' = 42(x+1)^5 - 42x^5.$$

$$P_7''(j) = 42(-j^2)^5 - 42j^5 = 42(-j - j^2) \neq 0.$$

then j is a double root of P_7 . since $P_7 \in \mathbb{R}[x]$
 $\bar{j} = j^2$ is also a root of multiplicity 2.

c) P_7 is of degree 6.

$$\text{In } \mathbb{Q}[x]; \quad P_7 = (x+1)x(x-j)^2(x-\bar{j})^2.$$

$$\text{In } \mathbb{R}[x]; \quad P_7 = (x+1)x((x-j)(x-\bar{j}))^2 \\ = (x+1)(x)(x^2+x+1)^2.$$

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Exercise 3: 1) a) $|G_{n,m}| = |\mathbb{Z}/n\mathbb{Z}| \times |\mathbb{Z}/m\mathbb{Z}| = n \times m$.

Then $|G_{2,3}| = 2 \times 3 = 6$.

$$G_{2,3} = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1}), (\bar{0}, \bar{2}), (\bar{1}, \bar{0}), (\bar{1}, \bar{1}), (\bar{1}, \bar{2})\}$$

b) It is easy to verify that $(\bar{1}, \bar{1})$ generates all the elements of $G_{2,3}$, then $(G_{2,3}, +)$ is cyclic.

$$G_{2,3} = \langle (\bar{1}, \bar{1}) \rangle. \quad G = \langle (\bar{1}, \bar{1}) \rangle = \langle -\bar{1}, -\bar{1} \rangle = \langle \bar{1}, \bar{2} \rangle.$$

c) Since 2 is a prime, it is enough to find an element of order 2 to get a subgroup of order 2. $2(\bar{1}, \bar{0}) = (\bar{0}, \bar{0})$. Then $H_1 = \langle (\bar{1}, \bar{0}) \rangle$ is a subgroup of order 2.

The same reasoning we get $H_2 = \langle (\bar{0}, \bar{1}) \rangle$ is a subgroup of order 3.

$\oplus$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{2})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{2})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{2})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{2})$
$(\bar{0}, \bar{2})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{2})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$
$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{2})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{2})$
$(\bar{1}, \bar{2})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{2})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{2})$	$(\bar{1}, \bar{1})$

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e) The zero divisors are $(\bar{0}, \bar{1}), (\bar{0}, \bar{2}), (\bar{1}, \bar{0})$.
The invertible elements are $(\bar{1}, \bar{1}), (\bar{1}, \bar{2})$.

$$2) S = \left\{ \frac{2m}{2n+1} \mid m, n \in \mathbb{Z} \right\}$$

a) S is a subring since $0 = \frac{2 \cdot 0}{2 \cdot 0 + 1} \in S$, then $S \neq \emptyset$.

$$\text{if } x = \frac{2m}{2n+1}, y = \frac{2m'}{2n'+1} \in \mathbb{Q}, \text{ then } x - y = \frac{2m(2n'+1) - 2m'(2n+1)}{(2n+1)(2n'+1)} \in S$$

$$\text{and } xy = \frac{2m \cdot 2m'}{(2n+1)(2n'+1)} = \frac{4mm'}{2(2nn' + n + n') + 1} \in S.$$

b) S is not an ideal of $\mathbb{Q}$, since $\frac{1}{2} \in \mathbb{Q}, 2 \in S$

$$\frac{1}{2} \times 2 = 1 \notin S.$$

c) S is not a subfield since $1 \notin S$.

3) since the laws $+$ and $\cdot$ are commutative

$x + y = xy$ gives $y + x = yx$ then if x is quasi-invertible with y as a quasi-inverse, y is quasi-invertible with x as quasi-inverse.

$$b) \quad x \in \text{Inv}(\mathbb{Z}) \Leftrightarrow x \in \mathbb{Z}, \exists y \in \mathbb{Z} \mid x+y=xy$$

$$\Leftrightarrow x \in \mathbb{Z}, \exists y \in \mathbb{Z} \mid x+y-xy=0$$

$$\Leftrightarrow x \in \mathbb{Z}, \exists y \in \mathbb{Z} \mid (x-1)(y-1)=1$$

$$\Leftrightarrow (x \neq 2 \wedge y=2) \vee (x=0 \wedge y=0).$$

$$\text{Then } \text{Inv}(\mathbb{Z}) = \{0, 2\}.$$

$$\text{Let } (\bar{a}, \bar{b}) \in \mathbb{Z}/_2\mathbb{Z} \times \mathbb{Z}/_3\mathbb{Z}, \quad (\bar{a}, \bar{b}) \in \text{Inv}(\mathbb{Z}/_2\mathbb{Z} \times \mathbb{Z}/_3\mathbb{Z}) \Leftrightarrow$$

$$\exists (\bar{a}', \bar{b}') \in \mathbb{Z}/_2\mathbb{Z} \times \mathbb{Z}/_3\mathbb{Z} \mid (\bar{a}, \bar{b}) \oplus (\bar{a}', \bar{b}') = (\bar{a}, \bar{b}) \otimes (\bar{a}', \bar{b}')$$

$$\Leftrightarrow \begin{cases} \bar{a} + \bar{a}' = \bar{a} \bar{a}' & \text{in } \mathbb{Z}/_2\mathbb{Z} \\ \bar{b} + \bar{b}' = \bar{b} \bar{b}' & \text{in } \mathbb{Z}/_3\mathbb{Z}. \end{cases}$$

$$\Leftrightarrow \begin{cases} \bar{a} = \bar{0}, \bar{a}' = \bar{0} \\ \bar{b} = \bar{0} \wedge \bar{b}' = \bar{0} \vee \bar{b} = \bar{2} \wedge \bar{b}' = \bar{2} \end{cases}$$

$$\text{Inv}(\mathbb{B}_{2,3}) = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{2})\}.$$

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$$\text{Inv}(S) = S \text{ ?}$$

$$\text{Let } x = \frac{2m}{2n+1}, m, n \in \mathbb{Z}, x \in S \quad \exists? x' = \frac{2m'}{2n'+1} \in S$$

$$\text{such that } x + x' = xx' \quad (1)$$

(1) is eq-valent to the linear equation

$$\text{in } m', n' \quad m(2n'+1) + m'(2n+1) = 2mm'$$

which has a solution $m' = -m$

$$n' = n - m,$$

then $x = \frac{2m}{2n+1}$ is quasi-invertible, its quasi-inverse

$$\text{is } x' = \frac{-2m}{2(n-m)+1}.$$

2) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ ^{or} a ring homomorphism, and y is the quasi-inverse of x .

We have $x + y = xy$, then

$$f(x+y) = f(xy), \text{ then } f(x) + f(y) = f(x)f(y).$$

Then $f(y)$ is the quasi-inverse of $f(x)$.

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4) a) on R , We define Δ by $x \Delta y = x + y - xy$.

c) Δ is commutative since $+$ and $\cdot$ are commutative.

ii) $\forall x, y, z \in R$, we have $(x \Delta y) \Delta z = (x + y - xy) \Delta z$

$$= (x + y - xy) + z - (x + y - xy)z$$

$$= x + y + z - xy - xz - yz + xyz \quad (1)$$

$$x \Delta (y \Delta z) = x \Delta (y + z - yz) = x + y + z - yz - x(y + z - yz)$$

$$= x + y + z - yz - xy - xz + xyz \quad (2)$$

(1) = (2), then Δ is associative.

iii) it is clear that $x + 0 - 0y = 0 + x - x0 = x$, then 0 is the neutral element of R with respect to Δ .

b) Let $x \in R$ and suppose that x admits two quasi-inverses x', x'' .

We have x' quasi-inverse of x

$$x + x' = xx' \text{ which is equivalent to}$$

$$x \Delta x' = 0. \text{ the same thing for } x''$$

$$x'' \Delta x = 0. \text{ Then}$$

$$\text{on one hand } (x'' \Delta x) \Delta x' = x'' \Delta (x \Delta x') = x'' \Delta 0 = x''$$

$$\text{On the other hand } (x'' \Delta x) \Delta x' = 0 \Delta x' = x', \text{ then}$$

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$$x = x'$$

c) suppose $a, b \in R$ quasi-invertible with a^* , b^* as quasi-inverses, thanks to the associativity and commutativity of Δ we have

$$(a \Delta b) \Delta (a^* \Delta b^*) = (a \Delta a^*) \Delta (b \Delta b^*)$$

$$= 0 \Delta 0 = 0.$$

then $a \Delta b$ is quasi-invertible with $(a \Delta b)^* = a^* \Delta b^*$.

d) $\text{Inv}(R) \neq \emptyset$ since $0 \in \text{Inv}(R)$.

Δ is closed in $\text{Inv}(R)$ thanks to question c).

Commutativity and associativity are clear since $\text{Inv}(R) \subset R$ and Δ is associative and commutative in R . $0 \in \text{Inv}(R)$ then 0 is neutral element

of $\text{Inv}(R)$ with respect Δ . Every element

of $\text{Inv}(R)$ is an element of R which has a quasi-inv i.e. $\exists y \in R$ $x \Delta y = 0$ this means

that x has an inverse with respect Δ .

Conclusion $(\text{Inv}(R), \Delta)$ is an abelian group.

e) It is clear that $\forall x \in R$ $x \Delta y = 0$ for some $y \in R$ is equivalent to $x + y = xy$, which means

That the quasi-invertible elements of $(R, +, \cdot)$ are exactly the invertible elements of (R, Δ) .

f) Let $x \in R$. x^2 quasi-invertible then there exists $y \in R$ such that $x^2 \Delta y = 0$, which gives $x^2 \Delta$

by noting that $x^2 = x \Delta (-x)$ we have
$$0 = x^2 \Delta y = (x \Delta (-x)) \Delta y = x \Delta (-x \Delta y) = 0$$

Then if x^2 is quasi-invertible with y as quasi-inverse, x is also quasi-invertible, with $(-x \Delta y)$ as quasi-inverse.

The converse is false since in $\mathbb{Z}$ 2 is quasi-invertible but 2^2 is not.

5) $(R, +, \cdot)$ is a Commutative ring with unity 1

$x \in R$ is quasi-invertible $\Leftrightarrow 1-x$ invertible in R ?

$x \in R$ quasi-invertible $\Leftrightarrow \exists y \in R$ such that $x+y-xy=0$

$\Leftrightarrow \exists y \in R, (x-1)(y-1)=1 \Leftrightarrow x-1$ invertible in R .

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b) 1 is not quasi-invertible since $1-1=0$ is not invertible in R (following a))

c) Let $x \in R$, $x^n = 0$ for some $n \in \mathbb{N}$. We have $1 = (1 - x^n) = (1 - x) \left(\sum_{k=0}^{n-1} x^k \right)$; then $(1 - x)$ is invertible, then x quasi-invertible (following a))

The Converse is false 2 is quasi-invertible in $\mathbb{Z}$ but 2 is not invertible.

d) Let $x \in R$, $x^2 = x$, $x \neq 0$. Suppose x quasi-invertible then there exists $y \in R$ such that $x + y - xy = 0$, We compose with

x we get $x^2 + xy - x^2y = 0$, then

$$x + xy - xy = 0 \quad (\text{since } x^2 = x)$$

then $x = 0$. Contradiction.